

Setting up Visual Studio Code Server

2020-12-10



This blog will explain how to set up a visual studio code server in an AWS instance (Ubuntu 18.04). The steps mentioned here are basically the commands I executed in my instance (launched using ami-0dba2cb6798deb6d8 64-bit x86) with security group rules allowing port 22,80 and 443 for all.

Note (October 2026): this post was written in 2020 and has not been re-tested. Ubuntu 18.04 and code-server 3.4.1 are long out of date, so use current versions and check the code-server documentation before following these steps. **Test1234** is only an example: choose your own strong password, and do not leave ports open to everyone on a real server.

Step 1: `sudo apt update`

Step 2: `wget https://github.com/cdr/code-server/releases/download/3.4.1/code-server`

Step 3: `tar xvzf code-server-3.4.1-linux-x86_64.tar.gz`

Step 4: `sudo mkdir -p ~/.config/code-server`

Step 5: `sudo nano ~/.config/code-server/config.yaml`

Paste in the config below, save, and check it with:

`sudo cat ~/.config/code-server/config.yaml`

Output:

```
bind-addr: 127.0.0.1:8080
auth: password
password: Test1234
cert: false
```

Step 6: `cd ~/code-server-3.4.1-linux-x86_64/bin`

Step 7: `./code-server`

Output:

```
info Using config file ~/.config/code-server/config.yaml
info Using user-data-dir ~/.local/share/code-server
info code-server 3.4.1 48f7c2724827e526eeaa6c2c151c520f48a61259
info HTTP server listening on http://127.0.0.1:8080
info     - Using password from ~/.config/code-server/config.yaml
info     - To disable use `--auth none`
info     - Not serving HTTPS
```

[Ctrl+C to stop this service]

Step 8: `sudo nano /lib/systemd/system/code-server.service`

Paste in the unit file below, save, and check it with:

```
sudo cat /lib/systemd/system/code-server.service
```

Output:

```
[Unit]
```

```
Description=code-server
```

```
After=nginx.service
```

```
[Service]
```

```
Type=simple
```

```
Environment=PASSWORD=Test1234
```

```
ExecStart=/home/ubuntu/code-server-3.4.1-linux-x86_64/bin/code-server --bind-addr=127.0.0.1:8080
```

```
Restart=always
```

```
[Install]
```

```
WantedBy=multi-user.target
```

Step 9: `sudo mkdir /var/lib/code-server`

Step 10: `sudo systemctl daemon-reload`

Step 11: `sudo systemctl start code-server`

Step 12: `sudo systemctl enable code-server`

Output:

Created symlink /etc/systemd/system/multi-user.target.wants/code-server.service

Step 13: `sudo systemctl status code-server`

Output:

```
• code-server.service - code-server
  Loaded: loaded (/lib/systemd/system/code-server.service; enabled; vendor pr
  Active: active (running) since Wed 2020-10-07 03:06:01 UTC; 22s ago
  Main PID: 2209 (node)
    Tasks: 22 (limit: 1164)
   Memory: 23.8M
    CGroup: /system.slice/code-server.service
            |-2209 /home/ubuntu/code-server-3.4.1-linux-x86_64/lib/node /home/u
            |-2237 /home/ubuntu/code-server-3.4.1-linux-x86_64/lib/node /home/u

Oct 07 03:06:01 ip-172-31-18-5 systemd[1]: Started code-server.
Oct 07 03:06:01 ip-172-31-18-5 code-server[2209]: info Wrote default config fil
Oct 07 03:06:01 ip-172-31-18-5 code-server[2209]: info Using config file ~/.con
Oct 07 03:06:01 ip-172-31-18-5 code-server[2237]: info Using user-data-dir /var
Oct 07 03:06:01 ip-172-31-18-5 code-server[2237]: info code-server 3.4.1 48f7c2
Oct 07 03:06:01 ip-172-31-18-5 code-server[2237]: info HTTP server listening on
Oct 07 03:06:01 ip-172-31-18-5 code-server[2237]: info - Using password fro
Oct 07 03:06:01 ip-172-31-18-5 code-server[2237]: info - To disable use `--
Oct 07 03:06:01 ip-172-31-18-5 code-server[2237]: info - Not serving HTTPS
```

Step 14: `sudo apt install nginx -y`

Step 15: `sudo nano /etc/nginx/sites-available/code-server.conf`

Paste in the server block below, save, and check it with:

`sudo cat /etc/nginx/sites-available/code-server.conf`

Output:

```
server {
    listen 80;
    listen [::]:80;
    server_name 54.XXX.XX.253;
    location / {
```

```
    proxy_pass http://localhost:8080/;
    proxy_set_header Host $host;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection upgrade;
    proxy_set_header Accept-Encoding gzip;
}
}
```

[*NOTE*: Remember to change “54.XXX.XX.253” to the public IP of your instance]

Step 16: `cd /etc/nginx/sites-available/`

Step 17: (OPTIONAL) `sudo mv default BackUp_default`

Step 18: `sudo rm /etc/nginx/sites-enabled/default`

Step 19: `sudo ln -s /etc/nginx/sites-available/code-server.conf /etc/nginx/sites-enabled/code-server.conf`

Step 20: `sudo systemctl start nginx`

Step 21: `sudo systemctl enable nginx`

Output:

Synchronizing state of nginx.service with SysV service script with /lib/systemd/
Executing: /lib/systemd/systemd-sysv-install enable nginx

Step 22: `sudo systemctl status nginx`

Output:

```
• nginx.service - A high performance web server and a reverse proxy server
  Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset:
  Active: active (running) since Wed 2020-10-07 03:06:47 UTC; 3min 12s ago
    Docs: man:nginx(8)
  Main PID: 2696 (nginx)
    Tasks: 2 (limit: 1164)
  Memory: 6.6M
  CGroup: /system.slice/nginx.service
          |-2696 nginx: master process /usr/sbin/nginx -g daemon on; master_p
          |-2697 nginx: worker process
```

Oct 07 03:06:46 ip-172-31-18-5 systemd[1]: Starting A high performance web server

Oct 07 03:06:47 ip-172-31-18-5 systemd[1]: Started A high performance web server

Step 23: `sudo systemctl daemon-reload`

Step 24: Refresh both the services by starting and stopping them. You can follow the commands below in order:

```
sudo systemctl stop nginx
sudo systemctl stop code-server
sudo systemctl daemon-reload
sudo systemctl start code-server
sudo systemctl start nginx
sudo systemctl enable code-server
sudo systemctl enable nginx
sudo systemctl status nginx # To check if nginx is running
sudo systemctl status code-server # To check if the code-server is active
```

Step 25: Open browser and try accessing the Public IP of the instance

Then, the VS code server should work now.